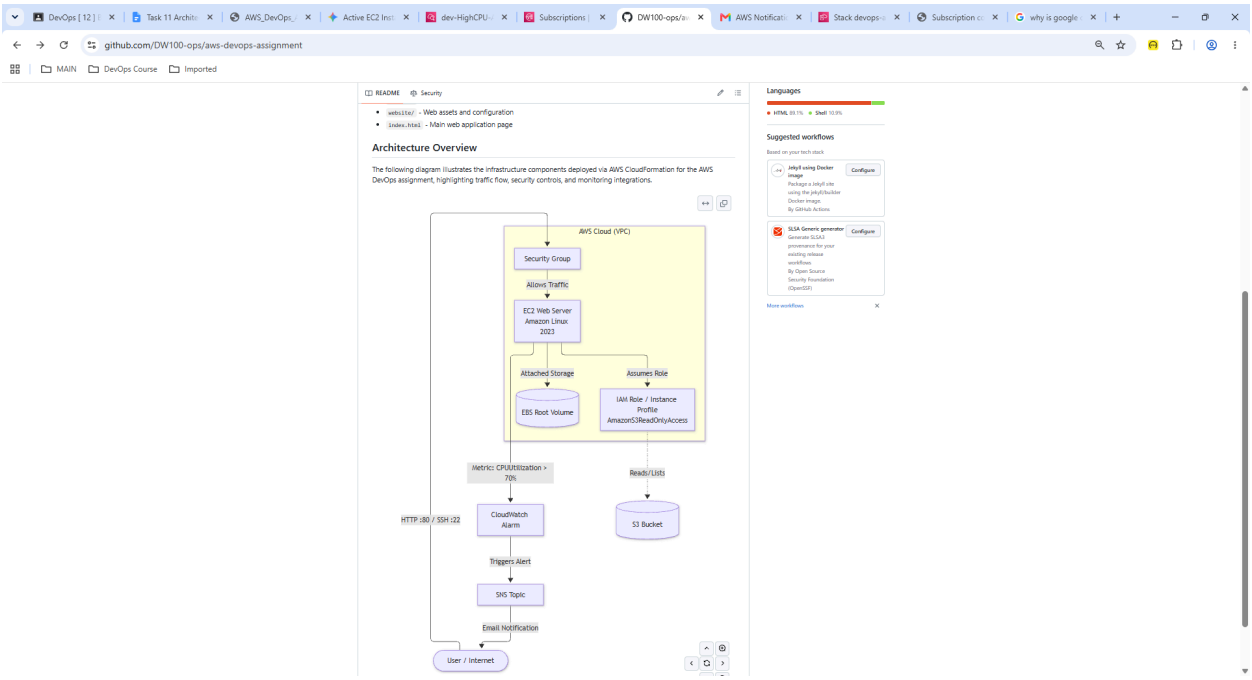


Component Breakdown & Data Flow:

1. **Internet / User:** Originates public HTTP requests to the web server and receives email notifications.
2. **Security Group (`WebServerSecurityGroup`):** Acts as a virtual firewall controlling inbound traffic on Port 80 (HTTP) and Port 22 (SSH).
3. **EC2 Web Server (`WebServerInstance`):** Hosts the Apache web server application serving the assignment index page.
4. **EBS Volume:** Provides persistent block storage for the EC2 operating system and file system.
5. **IAM Role & Instance Profile (`WebServerRole`):** Grants secure, credentials-free read access to AWS services (specifically S3).
6. **S3 Bucket (`MyS3Bucket`):** CloudFormation-managed object storage bucket with versioning enabled.
7. **CloudWatch Alarm (`CPUUtilizationAlarm`):** Continuously monitors EC2 CPU metrics and breaches if usage exceeds 70% over a 5-minute evaluation period.
8. **SNS Topic (`AlarmTopic`):** Broadcasts alarm events to subscribed administrative endpoints (email).